

Lab 3: Neuro - Measurement Station 2

Grego & Peti

October 30, 2025

Task 1: Series Resistor Measurement

$R_s = 2.17 \text{ k}\Omega$ (multimeter measurement), R_{smd} nominal: $4.701 \text{ k}\Omega$. Assembled circuit with R_s in series with R_{smd} . R_{smd} measured: $4.663 \text{ k}\Omega$. Value close to nominal, circuit validated. Code and circuit functioning.

Visuals

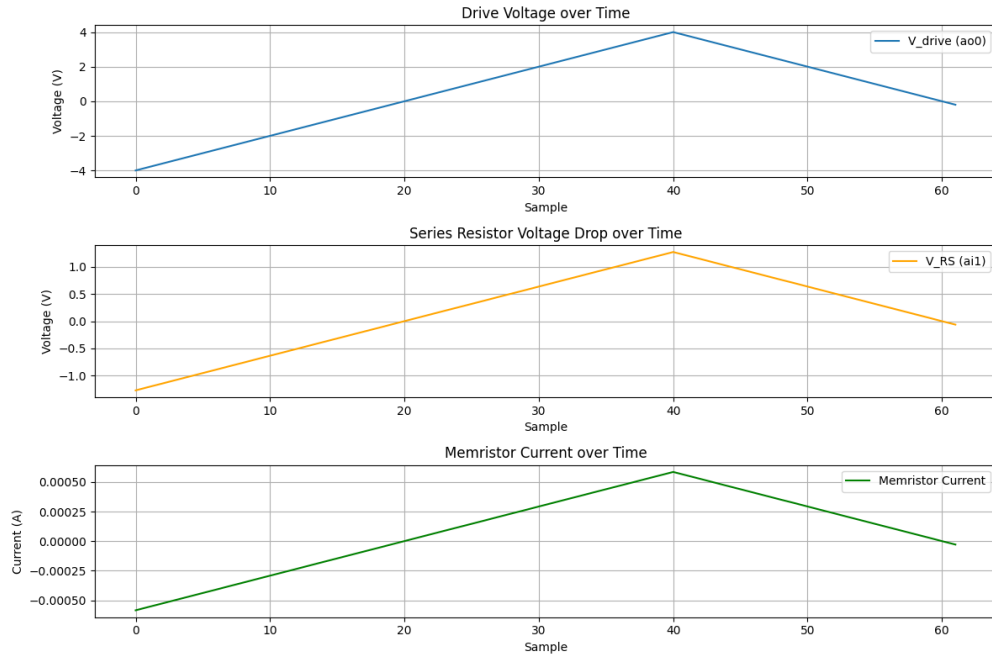


Figure 1: Task 1: Time Series Plot

Task 2: I(V) Curve Measurement

Observed expected I(V) curve with characteristic jumps. Using the row1 col3 memristor on the sample.

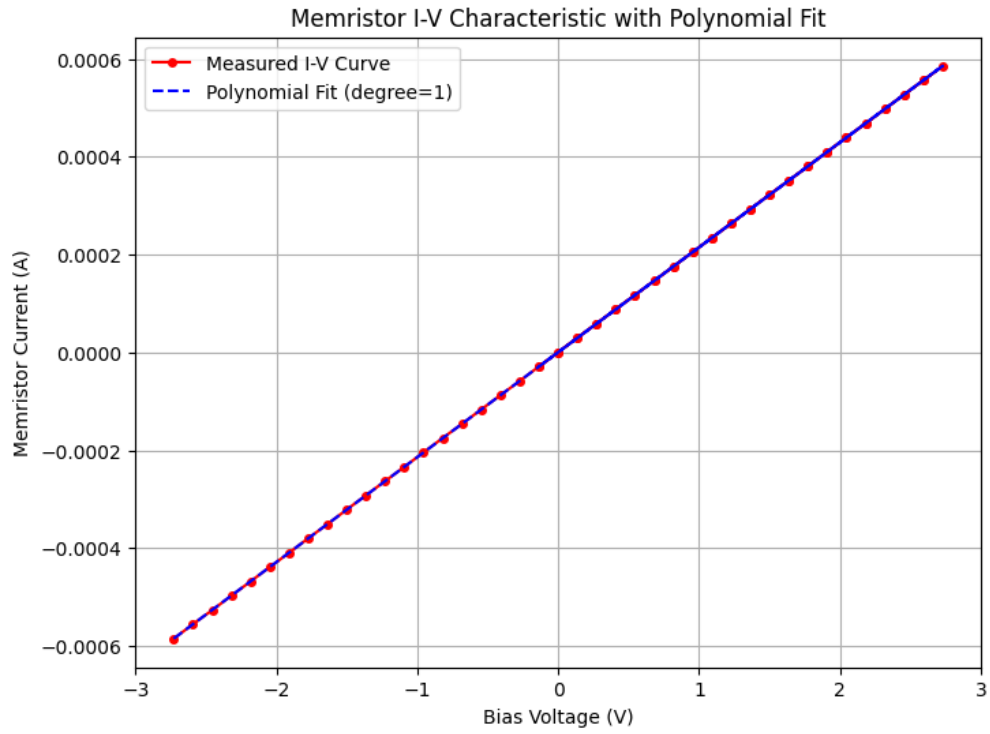


Figure 2: Task 1: Fit Plot

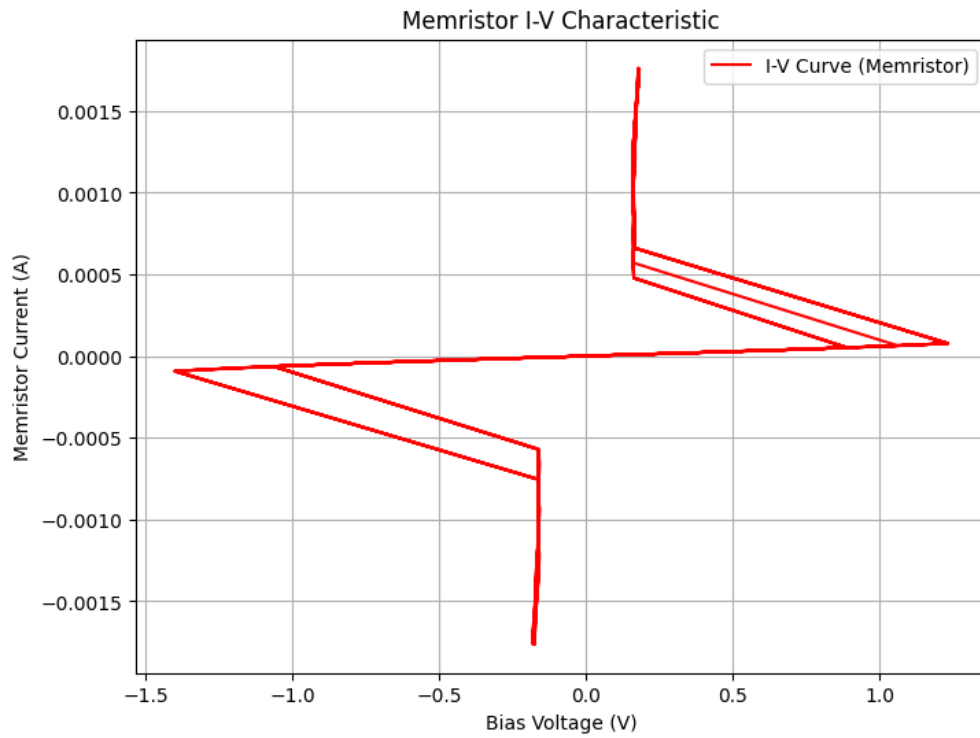


Figure 3: Task 2: Memristor I(V) Curve

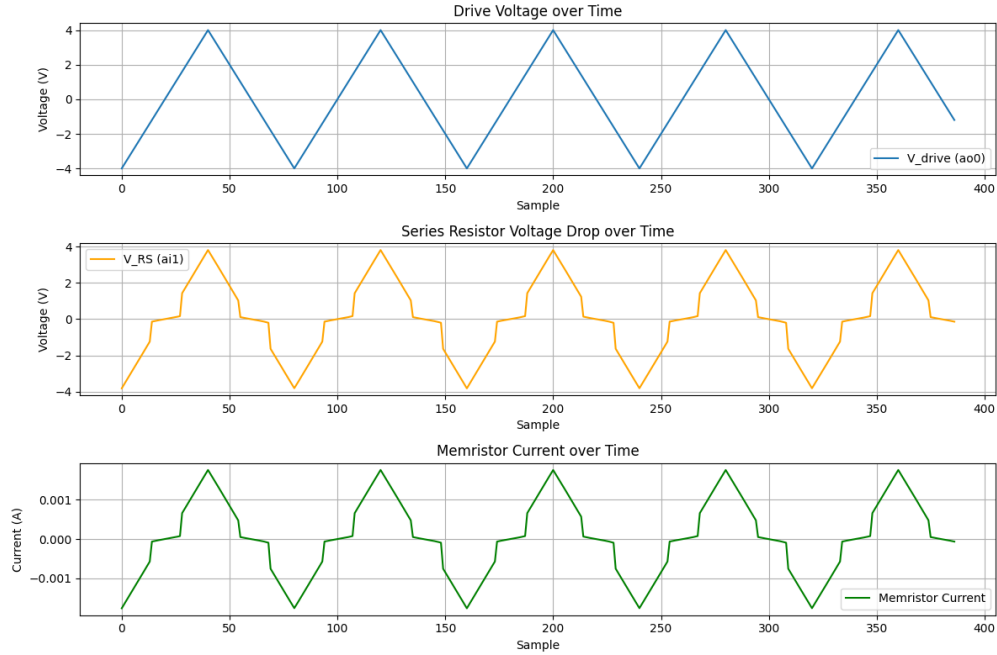


Figure 4: Task 2: Three Plots

Visuals

Task 3: $R(T)$ Characteristics Measurement Setup

User interface, VISA/SCPI control, and data acquisition setup completed for $R(T)$ measurements. Calculations for sample resistance and temperature implemented. Data logging and plotting updated. Power supply ON/OFF controls added. Remember to update the 'PeltierPowerSupplyVisaResource'.